

Yunseo Choi

Arachi Yoichi

Regression Analysis for Social Sciences 1

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Model that focuses on Screen Time and Sleep to Estimate Mental Health Score

1. Introduction

It is a common belief that longer screen time has a negative effect on mental health. However, when I was in a boarding school during the corona virus, I was able to stay close with my peers, and realized that some people seemed to have a stable mental health even with high screen time. This made me want to investigate how much effect screen time has on mental health, and what other factors can contribute to mental health. In Wang and their colleague's paper, he investigates the impact sleep status and screen time has on mental health. Through their study, they found out that "screen time was associated with depression and anxiety" while there was "no significant relationship between screen time and loneliness" (Wang et al). This again strengthens the idea that screen time has a substantial effect on some of the symptoms of mental health. In addition to this, they identified that sleep also has an impact on mental health as it "partially mediated" the effect of screen time on mental health, "accounting for 13.73% and 19.68% of the total effects" (Wang et al). Because of this, my work investigates the weight of sleep and screen time on mental health. It also investigates the interaction of gender and screen time as Chalermchutidej's research team stated "time spent on social media had a stronger relationship with emotional distress among female rather than male." By majorly focusing on the impact of screen time and sleep, and incorporating some demographic details of the people, the model attempts to predict the mental health score of the people.

2. Model and Describe Your Approach

The model I have created is the following:

Mental_Health_Score=

$$10.020 - 0.028 \cdot \text{Gender_Male} + 2.009 \cdot \text{Academic_Level_Graduate} + 1.787 \cdot \text{Academic_Level_Undergraduate} - 0.436 \cdot \text{Addicted_Score} + 0.225 \cdot \text{Relationship_Status_Complicated} + 0.331 \cdot \text{Relationship_Status_Single} - 0.054 \cdot \text{Avg_Daily_Usage_Hours} - 0.637 \cdot \log(\text{Sleep_Hours_Per_Night}) - 0.228 \cdot (\text{Academic_Level_Graduate} \times \text{Addicted_Score}) - 0.204 \cdot (\text{Academic_Level_Undergraduate} \times \text{Addicted_Score}) - 0.065 \cdot (\text{Relationship_Status_Single} \times \text{Avg_Daily_Usage_Hours}) - 0.043 \cdot (\text{Relationship_Status_Complicated} \times \text{Avg_Daily_Usage_Hours}) + \varepsilon$$

Before creating a model, dummy variables were created for categorical variables of Gender, Academic_Level, Affects_Academic_Performance, Relationship_Status. Then by using VIF test, variables that show high multicollinearity have been removed. For example, Addicted_Score had a high VIF score in the model that contains Conflicts_Over_Social_Media and Avg_Daily_Usage_Hours. The three of them had high positive correlation with each variable, minimum among them being 0.805, so F test was used to identify restricting which variable would create a better model. A model with restricted value of "Conflicts_Over_Social_Media=0" had a low F value, showing that restricted model performs better than the original model, so Conflicts_Over_Social_Media was removed.

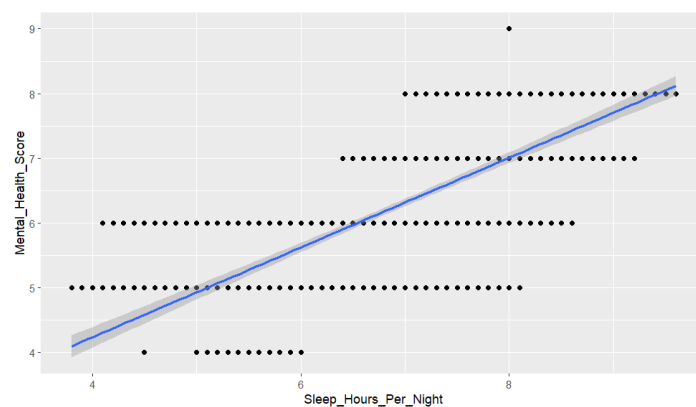
The process of checking multicollinearity among the variables and removing less powerful variables were repeated until we got the following model

```
model14 <- lm(Mental_Health_Score ~ Gender_Male+Academic_Level_Graduate+Academic_Level_Undergraduate+
  Relationship_Status_Complicated + Relationship_Status_Single+Avg_Daily_Usage_Hours +
  Affects_Academic_Performance_Yes+Sleep_Hours_Per_Night+Addicted_Score, data = media)
```

Afterwards, models with different interaction terms have been tested. As mentioned in the introduction, a model including interaction terms between Gender and Avg_Daily_Usage_Hours has been tested, but when the interaction term was restricted to 0, the F test yielded a low value showing that the restricted model is better than the non-restricted model, leading to the exclusion of the interaction term. By repeating this process for other interaction terms, we got the following model:

```
model18 <- lm(Mental_Health_Score ~ Gender_Male+Academic_Level_Graduate+Academic_Level_Undergraduate+
  Academic_Level_Graduate*Addicted_Score+Academic_Level_Undergraduate*Addicted_Score+
  Relationship_Status_Complicated + Relationship_Status_Single+Avg_Daily_Usage_Hours +
  Relationship_Status_Single*Avg_Daily_Usage_Hours+Relationship_Status_Complicated*Avg_Daily_Usage_Hours+
  Sleep_Hours_Per_Night+Addicted_Score, data = media)
```

Plots of explanatory of variable and the explained variable were created to see if any of them deviate from a linear relationship.



Sleep_Hours_Per_Night seem to have a logistic relationship with the explained variable, so another model with $\log(\text{Sleep_Hours_Per_Night})$ was tested to see if it yielded a better R^2 score.

Observations	705	705
R2	0.900	0.901
Adjusted R2	0.898	0.899
Residual Std. Error (df = 692)	0.353	0.351
F Statistic (df = 12; 692)	516.839***	524.513***
Note:	*p<0.1; **p<0.05; ***p<0.01	

The left column of the chart shows the analysis result for the model8, and the right column of the chart shows the analysis result for the model9, one that replaces Sleep_Hours_Per_Night with $\log(\text{Sleep_Hours_Per_Night})$ from model8. Model9 had a better R^2 score and a higher F statistic, so I concluded it as the final model.

3. Empirical Analysis

The data for the model was collected from Kaggle, and the following descriptions are the exact Kaggle descriptions of how they were collected

Age	student's age in completed years at the time of the survey
Gender	student's self-reported gender, recorded as "Male" or "Female"
Academic_Level	The highest level of education the respondent is currently enrolled in, with categories: "High School," "Undergraduate," or "Graduate," facilitating stratified analyses by academic stage.
Avg_Daily_Usage_Hours	The average number of hours per day the student spends on social media platforms, calculated from self-reported weekday and weekend usage estimates.
Affects_Academic_Performance	A binary indicator ("Yes"/"No") reflecting whether the student perceives their social media use as having a negative impact on their academic performance.
Sleep_Hours_Per_Night	The respondent's average nightly sleep duration in hours, provided to investigate correlations between screen time and sleep quality/quantity.
Relationship_Status	The student's current romantic relationship status, categorized as "Single," "In Relationship," or "Complicated," to explore social media's impact on interpersonal dynamics.
Conflicts_Over_Social_Media	The number of arguments or disagreements the student reports having had with family, friends, or partners due to their social media use, serving as a proxy for social friction.
Addicted_Score	A composite score from 1 (low addiction) to 10 (high addiction) based on a standardized survey scale (e.g., Bergen Social Media Addiction Scale), quantifying the degree of problematic usage.
Mental_Health_Score	A self-rated integer from 1 (poor) to 10 (excellent) indicating overall mental well-being, allowing assessment of potential associations with social media habits.

Descriptive statistics of the variables are as follows:

Student_ID	Age	Gender	Academic_Level	Country	Avg_Daily_Usage_Hours
Min. : 1	Min. :18.00	Length:705	Length:705	Length:705	Min. :1.500
1st Qu.:177	1st Qu.:19.00	Class :character	Class :character	Class :character	1st Qu.:4.100
Median :353	Median :21.00	Mode :character	Mode :character	Mode :character	Median :4.800
Mean :353	Mean :20.66				Mean :4.919
3rd Qu.:529	3rd Qu.:22.00				3rd Qu.:5.800
Max. :705	Max. :24.00				Max. :8.500
Most_Used_Platform	Affects_Academic_Performance	Sleep_Hours_Per_Night	Mental_Health_Score	Relationship_Status	
Length:705	Length:705	Min. :3.800	Min. :4.000	Length:705	
Class :character	Class :character	1st Qu.:6.000	1st Qu.:5.000	Class :character	
Mode :character	Mode :character	Median :6.900	Median :6.000	Mode :character	
		Mean :6.869	Mean :6.227		
		3rd Qu.:7.700	3rd Qu.:7.000		
		Max. :9.600	Max. :9.000		
Conflicts_Over_Social_Media	Addicted_Score				
Min. :0.00	Min. :2.000				
1st Qu.:2.00	1st Qu.:5.000				
Median :3.00	Median :7.000				
Mean :2.85	Mean :6.437				
3rd Qu.:4.00	3rd Qu.:8.000				
Max. :5.00	Max. :9.000				

The model I have created has been estimated as the following:

Mental_Health_Score=

$$10.020 - 0.028 \cdot \text{Gender_Male} + 2.009 \cdot \text{Academic_Level_Graduate} + 1.787 \cdot \text{Academic_Level_Undergraduate} - 0.436 \cdot \text{Addicted_Score} + 0.225 \cdot \text{Relationship_Status_Complicated} + 0.331 \cdot \text{Relationship_Status_Single} - 0.054 \cdot \text{Avg_Daily_Usage_Hours} - 0.637 \cdot \log(\text{Sleep_Hours_Per_Night}) - 0.228 \cdot (\text{Academic_Level_Graduate} \times \text{Addicted_Score})$$

dicted_Score)-0.204·(Academic_Level_Undergraduate×Addicted_Score)-0.065·(Relationship_Status_Single×Avg_Daily_Usage_Hours)-0.043·(Relationship_Status_Complicated×Avg_Daily_Usage_Hours)+ ε

The chart below shows the measures to show the model's performance. The model has a high adjusted R² of 0.899, with a significant F statistic of 524.513.

Observations	705
R2	0.901
Adjusted R2	0.899
Residual Std. Error	0.351 (df = 692)
F Statistic	524.513*** (df = 12; 692)
=====	
Note:	*p<0.1; **p<0.05; ***p<0.01

In the table below, significant coefficients are represented in stars, with the increasing number of stars showing increasing significance. Most of the coefficients had a significance less than 0.05, while Gender's coefficient was marked insignificant. However, Gender is still included in the model because a lot of studies separate the gender groups when investigating for mental health. Also, the R² value didn't differ a lot when it was included.

	Mental_Health_Score

Gender_Male	-0.028 (0.036)
Academic_Level_Graduate	2.009*** (0.512)
Academic_Level_Undergraduate	1.787*** (0.513)
Addicted_Score	-0.436*** (0.064)
Relationship_Status_Complicated	0.225 (0.278)
Relationship_Status_Single	0.331*** (0.114)
Avg_Daily_Usage_Hours	-0.054** (0.026)
log(Sleep_Hours_Per_Night)	-0.637*** (0.144)
Academic_Level_Graduate:Addicted_Score	-0.228*** (0.064)
Academic_Level_Undergraduate:Addicted_Score	-0.204*** (0.064)
Relationship_Status_Single:Avg_Daily_Usage_Hours	-0.065*** (0.022)
Relationship_Status_Complicated:Avg_Daily_Usage_Hours	-0.043 (0.058)
Constant	10.020*** (0.605)

4. Robustness Check

Measuring mental health and screen time is difficult as it requires the values to be self-reported. This can be problematic because a bias can be created when people would answer in a way to benefit their image, rather than reporting the truth. In our study, some of the measures such as `Mental_Health_Score`, `Avg_Daily_Usage_Hours` are self-reported, and if the values are biased, the model that is estimated would lose credibility.

5. Conclusion

In conclusion, our model mainly focusses on people's use of screen and their length of sleep to estimate Mental Health Score. The model is further refined by including demographic details such as relationship status and academic level, and the interaction effects among the meaningful variables. The model can help to identify people who are in danger of mental health, by gathering their information of screen time, sleeping hours, and other demographic details. Future research can be conducted to investigate the reason behind gender not being a significant estimator as our common belief.

6. Works Cited

Chalermchutidej, Wikanda et al. "Sleep, social media use and mental health in female adolescents aged 12 to 18 years old during the COVID-19 pandemic." *BMC pediatrics* vol. 23,1 398. 14 Aug. 2023, doi:10.1186/s12887-023-04218-4

Wang, Wei et al. "Relationship between mental health, sleep status and screen time among university students during the COVID-19 pandemic: a cross-sectional study." *BMJ open* vol. 13,12 e073347. 9 Dec. 2023, doi:10.1136/bmjopen-2023-073347